

Survey of Discrete Distributions of Order k and Related Distributions

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The distribution of the number of trials until the first occurrence of consecutive k successes in Bernoulli trials with success probability p is called a geometric distribution of order k . This definition is due to Philippou, Georgiou and Philippou (1983). Though the distribution seems to have been interested since De Moivre (1667-1754)'s works (Todhunter (1865), Feller (1968), Johnson, Kotz and Kemp (1992, pp. 426-432) and Balakrishnan and Koutras (2002)), Philippou et al. (1983)'s contribution in this field seems to be very important. Since Philippou et al. (1983) called the distribution a geometric distribution of order k and defined a Poisson distribution of order k and a negative binomial distribution of order k , attention has been paid to inter-relationships among the so called discrete distributions of order k and exact distribution theory has been developed extensively by many researchers.

From early 80's, reliability theory of the consecutive- k -out-of- n :F systems began to be studied (Kontoleon (1980), Chiang and Niu (1981) and Derman, Lieberman and Ross (1982)), and a large number of studies have been made as relationships between the reliability of the system and discrete distributions of order k have been clear (e.g. Lambiris and Papastavridis (1985), Fu (1985, 1986a, 1986b), Fu and Beihua (1987), Aki (1985), Hirano (1986),

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Philippou (1986), Papastavridis (1987), Aki and Hirano (1988, 1989, 1996, 1997), Chrysaphinou and Papastavridis (1990c), Fu and Koutras (1994a,b), Godbole (1990a, 1993), Griffith (1994), Hirano and Aki (1993), Koutras, Papadopoulos and Papastavridis (1994), Charalambides (1994) and Makri and Philippou (1994)).

Various modifications have been made also on the underlying sequence, e.g. replacing the Bernoulli trials for other random sequences such as some urn models, a binary sequence of order k (Aki (1985), Hirano and Aki (1987), Aki (1992), Aki and Hirano (1994, 1995), Dhar and Jiang (1995), Balakrishnan (1997) and Balakrishnan and Koutras (1998)), Markov dependent trials and higher order Markov dependent trials. Consequently, the class of discrete distributions of order k has been extended and become fruitful. Moreover, some other kinds of extensions are to be examined; adoption of different ways of counting the number of runs, constructing a multivariate version of the class of the distributions (e.g. Johnson, Kotz and Balakrishnan (1997, Chapter 42)), and introduction of various waiting time problems.

1 Discrete distributions of order k

First, we will pay attention to the following relationships among the binomial distribution $B(n, p)$, the negative binomial distribution $NB(r, p)$, the geometric distribution $G(p)$, the Poisson distribution $P(\lambda)$, and the logarithmic series distribution $LS(p)$. The binomial distribution is related to the negative binomial distribution by the notion of *inverse sampling*, i.e., $\sum_{x \leq n} NB(r, p; x)$

$= \sum_{x \geq r} B(n, p; x)$. By setting $r = 1$ in the negative binomial distribution $NB(r, p)$, we get the geometric distribution $G(p)$. The r -th convolution power of the geometric distribution $G(p)$ becomes the negative binomial distribution $NB(r, p)$. Setting $p \rightarrow 1$, $r \rightarrow \infty$ and $r(1 - p) \rightarrow \lambda$ in the negative binomial distribution $NB(r, p)$ yields the Poisson distribution $P(\lambda)$. By com-

pounding a Poisson distribution, we can get a negative binomial distribution; the conditional distribution of X given $Y = y$ is a Poisson distribution $P(\lambda y)$, and Y follows the gamma distribution with density $\frac{\alpha^r}{\Gamma(r)} y^{r-1} e^{-\alpha y}$, then X is distributed as $NB(r, p)$, where $q = 1 - p = \frac{\lambda}{\lambda + \alpha}$. When X follows the negative binomial distribution $NB(r, p)$, the conditional probability $P(X = x | X > 0)$ converges to $LS(p; x)$ as $r \rightarrow 0$. By generalizing the Poisson distribution $P(-r \log p)$ with the generalizer $LS(p)$, we obtain the negative binomial distribution $NB(r, p)$.

Note that the geometric $G(p)$, the binomial $B(n, p)$, and the negative binomial $NB(r, p)$ distributions are based on independent Bernoulli trials with success probability p . By changing the role of a success for that of a success-run of length k , we can obtain naturally a geometric distribution of order k , $G_k(p)$, a binomial distribution of order k , $B_k(n, p)$, and a negative binomial distribution of order k , $NB_k(r, p)$. By defining a Poisson distribution of order k , $P_k(p)$ and a logarithmic series distribution of order k , $LS_k(p)$ appropriately, corresponding relationships to the above among distributions of order k can be obtained (Philippou et al. (1983) and Aki, Kuboki and Hirano (1984)).

Aki (1985) defined a binary sequence of order k as an extension of a sequence of Bernoulli trials. Based on the sequence, the extended geometric distribution of order k , $EG_k(p_1, \dots, p_k)$, the extended binomial distribution of order k , $EB_k(n, p_1, \dots, p_k)$, the negative binomial distribution of order k , $ENB(r, p_1, \dots, p_k)$, the extended Poisson distribution of order k , $EP_k(\lambda_1, \dots, \lambda_k)$, and the extended logarithmic series distribution of order k , $ELS_k(p_1, \dots, p_k)$ are defined and still retain the corresponding relationships to the above. These distributions are equivalent to the multiparameter distributions of order k defined by Philippou (1988), such as $NB_k(r, q_1, \dots, q_k)$, $P_k(\lambda_1, \dots, \lambda_k)$ and $LS_k(q_1, \dots, q_k)$ in the sense that they can be obtained by changing the parameters appropriately from the corresponding distributions. We also note that the cluster negative binomial distribution defined by

Xekalaki and Panaretos (1989) based on an urn model with replacement is equivalent to the extended negative binomial distribution of order k ; a genesis scheme of the multiparameter (or cluster) negative binomial distribution of order k ($NB_k(r; q_1, \dots, q_k)$) is the following: An urn contains balls bearing the letters $F_1, \dots, F_k$ and $S(\equiv S_0)$ with respective proportions $q_1, \dots, q_k$ and p ($0 < q_i < 1$ for $1 \leq i \leq k$, $q_1 + \dots + q_k < 1$ and $q_1 + \dots + q_k + p = 1$). Balls are drawn from the urn with replacement until r balls ($r \geq 1$) bearing the letter S appear. Then the distribution of the sum of indices of the letters on the balls drawn is $NB_k(r; q_1, \dots, q_k)$. The sampling scheme called a cluster sampling can be seen as the $\{S, F\}$ -trials until the r -th occurrence of consecutive k successes by replacing the event F_i for $\underbrace{S \cdots S}_{i-1} F$, $i = 1, \dots, k$, and S_0 for $\underbrace{S \cdots S}_k$.

A binary sequence $\{X_i\}$ is said to be a *binary sequence of order k* if there exist a positive integer k and k real numbers $0 < p_1, p_2, \dots, p_k < 1$ such that
(1) $X_0 = 0$ almost surely and
(2) $P(X_n = 1 | X_0 = x_0, X_1 = x_1, \dots, X_{n-1} = x_{n-1}) = p_j$
is satisfied for any positive integer n , where $j = r - [(r-1)/k] \cdot k$ and r is the smallest positive integer which satisfies $x_{n-r} = 0$.

Then, by setting $q_1 = Q_1$, $q_i = P_1 \cdots P_{i-1} Q_i$ ($2 \leq i \leq k$), the relation $NB_k(r; q_1, \dots, q_k) = \overline{ENB}_k(r, P_1, \dots, P_k)$ holds. Similarly, we have the relation between the two logarithmic series distributions, $LS_k(q_1, \dots, q_k) = ELS_k(P_1, \dots, P_k)$.

When we consider a consecutive- k -out-of- n :F system based on the above binary sequence of order k , the system coincides with the consecutive- k -out-of- n : F system with $(k-1)$ -step Markov dependence introduced by Fu (1986), because the system does not work if consecutive k components fail. We have to note that we can consider m -consecutive- k -out-of- n :F system based on

the binary sequence of order k , whereas $(k - 1)$ -step Markov dependence no longer holds if $m > 1$.

Some various attempts to replace independent sequences with other dependent sequences have been made. Panaretos and Xekalaki (1986b) defined a hypergeometric distribution of order k and a negative hypergeometric distribution of order k based on some urn models which provide interesting sampling schemes. More general sampling schemes were introduced: We consider an urn that contains a white balls and b black balls. Assume that n balls are randomly drawn, one at a time and that each ball is returned to the urn along with c additional balls of the same color before the next drawing. A distribution is said to be the Pólya distribution of order k , to be denoted by $Pol_k(a, b, c, n)$, if it is the distribution of the number of occurrences of consecutive k white balls. In particular, when $c = 1$, the Pólya distribution is said to be the negative hypergeometric distribution of order k and denoted by $NH_k(a, b, n)$. These distributions were studied by Aki and Hirano (1988), Philippou, Tripsiannis and Antzoulakos (1989), Panaretos and Xekalaki (1989) and Godbole (1990c). Many authors tried to investigate the distribution theory based on a Markov chain. Rajarshi (1974) derived the pgf of the distributions corresponding to the geometric distribution of order k and the binomial distribution of order k . Schwager (1983) studied distributions on runs based on (L -order) Markov chain and discussed the distribution of the number of overlapping runs until the n -th trial. Recently, Aki and Hirano (1993, 1994, 1995), Hirano and Aki (1993), Balasubramanian, Viveros and Balakrishnan (1993), Mohanty (1994), and Antzoulakos and Philippou (1997b) studied exact distributions of runs based on a Markov chain.

Here, we give a historical remark on dependent $\{0, 1\}$ -sequences briefly. Various extensions from independent trials to dependent trials have been attempted in various situations. In particular, the following works are related to Markov dependent trials, Edwards (1960), Gabriel (1959), Helgert (1970), Plots (1973) and Rajarshi (1974). Bahadur (1961) treated dependent $\{0, 1\}$ -

sequence of length n generally. Based on the work, Kupper and Haseman (1978), Altham (1978) and Ng (1989) treated dependence caused by a litter effect in the field of toxicology. Recently, a Markov dependence model has been studied in the field (Rudolfer (1990)).

Distributions of order k and extended distributions of order k are further extended to multivariate distributions by Philippou, Antzoulakos and Tripsiannis (1989), Philippou and Antzoulakos (1990), Philippou, Antzoulakos and Tripsiannis (1990), Antzoulakos and Philippou (1994) and Antzoulakos and Philippou (1997a).

1.1 Discrete distribution related to succession events

A geometric distribution of order k is a distribution of waiting time (number of trials) until the first occurrence of consecutive k successes in Bernoulli trials. A good deal of effort has been made on waiting times for the first occurrence of a given pattern or word in independent or dependent trials (e.g. Rajarshi (1974), Li (1980), Gerber and Li (1981), Chrysaphinou and Papastavridis (1990a,b), Chrysaphinou et al. (1994) and Móri (1991)). Ebneshahrashoob and Sobel (1990) considered two succession events, a success run of length k and a failure run of length r and derived the exact distribution of the sooner waiting time (the number of trials until the first occurrence of one of the two events) and that of the later waiting time (the number of trials until the first occurrence of the both events). Aki (1992) generalized this problem to the case of a sequence of independent nonnegative-integer-valued random variables and obtained the exact solution of the n -th waiting time problem. Aki and Hirano (1993) and Balasubramanian, Viveros and Balakrishnan (1993) solved the sooner and later problem in the sequence of a Markov chain. Ling (1992) and Sobel and Ebneshahrashoob (1992) developed the sooner and later problem in the case of a frequency quota. Let $X_1, X_2, \dots$ be a time-homogeneous $\{0, 1\}$ -valued Markov chain. Let F_0 be the event that l runs of "0" of length r occur and let F_1 be the event that

m runs of "1" of length k occur in the sequence $X_1, X_2, \dots$. Uchida and Aki (1995) obtained recurrence relations of the probability generating functions of the distributions of the waiting time for the sooner and later occurring events between F_0 and F_1 . The waiting time to finish a volleyball game is a special example of the distributions. Aki, Balakrishnan and Mohanty (1996) studied the sooner and later waiting time problems based on higher order Markov dependent trials. Aki (1997) studied the distribution of the number of occurrences of the sooner event until the first occurrence of the later event based on a two-state Markov dependent trials and generalized the problem to a multivariate case. Koutras (1997a) investigated relationships between the number of appearances of a pattern and the waiting time of the r -th occurrence of the pattern in terms of their generating functions. Koutras (1997b) developed a general technique for the evaluation of the exact distribution in a wide class of waiting time problems. Koutras and Alexandrou (1997b) investigated the sooner waiting time problems in a sequence of trinary trials. Considering some enumeration schemes of the number of runs, we can derive more fruitful results. Feller (1968, Chapter XIII) defined a way of counting the number of runs exactly of length k as counting the number from scratch every time a run occurs. For example, the sequence $SSS|SFSSS|SSS|F$ contains 3 success runs of length 3. Though Feller's enumeration scheme is convenient for applying renewal theory, some other enumeration schemes are also interested from the practical view point. In the classical literature, a "success run of length k " meant an uninterrupted sequence of either exactly k , or of at least k successes (Feller (1968)). In this way of counting the number of runs of length 3 (or more), the above example $SSSS|FSSSSSS|F$ contains 2 success runs of length 3 (or more). In Goldstein (1990) a Poisson approximation of the distribution of the number of success runs of length k or more until the n -th trial was proposed. Schwager (1983) and Ling (1988, 1989) studied the distributions on the number of overlapping runs of length k . In this enumeration scheme, $SSSSFSSSSSSSF$ contains 6 overlapping

success runs. Ling (1988) obtained a recurrence relation for the pgf of the number of overlapping success runs of length k until the n -th trial in independent trials with success probability p . The distribution is called the type II binomial distribution of order k ($B_k^{II}(n, p)$). Hirano et al. (1991) obtained the exact pgf of $B_k^{II}(n, p)$ and its probability function. Ling (1989) and Hirano et al. (1991) studied the distribution of the number of trials until the r -th overlapping success run occurs. The distribution is called the type III negative binomial distribution of order k ($NB_k^{III}(r, p)$). Hirano and Aki (1993) derived the pgf's of the distributions of the numbers of overlapping success runs of length k and success runs of length k or more until the n -th trial in the sequence of a $\{0, 1\}$ -valued Markov chain. We have to note here that the probability that a success run of length k does not occur coincides with each other for the above three enumeration schemes. When we investigate the distribution of runs by using a probability generating function, the distribution of the number of success runs of length k or more becomes surprisingly simple, and as a byproduct the elegant result of Lambiris and Papastavridis (1985) on the reliability of a consecutive- k -out-of- n :F system is derived simply and alternatively (without using enumerative combinatorics) (Hirano and Aki (1993)).

As we have seen, the reliability of a consecutive- k -out-of- n :F system can be obtained as an application of the distribution theory of runs (e.g. Hirano (1994) and Chao, Fu and Koutras (1995)). The results of Goldstein (1990) have application to analysis of DNA data.

Shmueli and Cohen (2000) applied exact probabilities of $G_k(p)$, $NB_k(r, p)$ and $B_k(n, p)$ to the US Military Standard switching rules in sampling inspection by attributes.

Further, we have the following important application called start-up demonstration tests; a start-up demonstration test is a mechanism by which a vender demonstrates to a customer the reliability of a equipment with regard to its starting. The vender repeats start-ups of the equipment, such as lawn

mowers, chain saws, water pumps and outboard motors until a specified number of consecutive successful start-ups are observed. The pioneering work of the problem is Hahn and Gage (1983). Since Hahn and Gage (1983) regarded the sequence of start-ups as an independent trials with success probability p for simplicity, the number of attempted start-ups until the item is accepted follows the geometric distribution of order k , where k is the required number of consecutive successful start-ups to achieve acceptance. Viveros and Balakrishnan (1993) studied statistical inference from start-up demonstration test data and introduced a Markov dependence model. Balakrishnan, Balasubramanian and Viveros (1995) introduced corrective action models and Balakrishnan, Mohanty and Aki (1997) gave the joint probability generating function of some random variables appearing in the Markov dependence model of the start-up demonstration test with corrective actions.

Though these results on consecutive systems and start-up demonstration tests have been developed independently from theory of distributions of order k , the results on statistical inference are related deeply to Philippou, Georgiou and Philippou (1983) and Aki and Hirano (1989). And the distribution of number of start-ups (or trials) based on Markov dependent trials is related to the results of Aki (1985), Aki and Hirano (1993), Balasubramanian, Viveros and Balakrishnan (1993), Aki and Hirano (1994, 1995), Mohanty (1994), Uchida and Aki (1995) and Aki, Balakrishnan and Mohanty (1996).

In addition to the above applications, we have to mention the moving window detection procedure for discrete data. A consecutive- k -out-of- r -from- n :F system is a sequence of n linearly-ordered components such that the system fails if and only if there are r consecutive components from which at least k are failed. Since it includes a consecutive- k -out-of- n :F system (in the case of $r = k$) and k -out-of- n :F system (in the case of $n = r$), the system is a very general one (Griffith (1986), Papastavridis and Sfakianakis (1991), Sfakianakis, Kounias and Hillaris (1992), Papastavridis and Koutras (1993)

and Cai (1994)). To calculate the reliability of the system, it suffices to derive the distribution of the waiting time for the first observation of the window (consecutive trials) of length r which has at least k failures in independent trials with success probability p . The problem is called the moving window detection procedure and has been studied in the theory of radar detection, time sharing systems and quality control (Greenberg (1970), Saperstein (1973), Nelson (1978), Mirstik (1978) and Glaz (1983)). The moving window detection procedures have been extended in more general situations and they are treated and studied as the scan statistics (Glaz (1989), Glaz and Naus (1991), Koutras and Alexandrou (1995) and Glaz and Balakrishnan (1998)). By putting together waiting time problems and the enumeration schemes of the number of runs, more interesting problems can be solved. For example, the following results were obtained. For simplicity, we consider only the case of Bernoulli trials. The distribution of the number of failures until the first occurrence of consecutive k successes is a geometric distribution (of order 1), $G(p^k)$. The number of successes until the first occurrence of consecutive k successes is a shifted geometric distribution of order $(k - 1)$, $G_{k-1}(p, k)$. Similarly, the number of overlapping success runs of length l until the first occurrence of consecutive k successes becomes a shifted geometric distribution of order $(k - l)$, $G_{k-l}(p, k - l + 1)$. Aki and Hirano (1994) studied the above results based on a Markov chain. Aki and Hirano (1995) derived the joint distribution of the numbers of failures, successes and success runs of length less than k until the first occurrence of consecutive k successes in a two-state Markov chain. Hirano, Aki and Uchida (1997) studied the problem based on the m -th order Markov dependent trials and showed that the distributions of the number of success runs of length l in the cases of $m \leq l < k$ and $m > l$ are completely different.

Beyond the above exact distribution theory, Poisson approximations of the distributions are also interested by many authors. Most results are derived by the Stein-Chen method (Arratia, Goldstein and Gordon (1989, 1990),

Goldstein (1990), Godbole (1990b, 1991, 1992, 1993), Godbole and Gornowicz (1992), Barbour, Holst and Janson (1992), Godbole and Schaffner (1993), Fu (1985, 1986a, 1986b, 1993), Papastavridis (1987), Wang (1993) and their references).

There are not so many papers which treat statistical inference of parameters of the above distributions, since the probability functions are too complicated. It is not so easy to obtain maximum likelihood estimate of a parameter of the complicated distribution like above even by using computers. We can mention Douglas (1955) and Shumway and Gurland (1960a, 1960b) for estimating parameters of some special generalized Poisson distributions such as Neymann type A contagious distribution, Poisson binomial distribution and Poisson Pascal distribution. Philippou et al. (1983) proposed moment estimation of the parameters of $G_k(p)$, $NB_k(p)$ and $P_k(\lambda)$. Aki and Hirano (1989) discussed methods for obtaining the maximum likelihood estimates of the parameters of discrete distributions of order k by using the Newton-Raphson method and recurrence relations of derivatives of probabilities of the distributions. After some numerical calculations, they also showed that the asymptotic efficiencies of the moment estimators are very close to one for given parameters which belong to relatively wide ranges. Aki (1990) treated a method for obtaining the m.l.e. of a two-parameter model; as an application simultaneous estimation of the parameters r and p in the negative binomial distribution of order k . Balakrishnan, Balasubramanian and Viveros (1995) discussed statistical inference for the probability p of a successful start-up based on demonstration test data. Aki and Hirano (1996) gave the distribution of the lifetime of consecutive- k -out-of- n :F systems with i.i.d. components as a mixture of the distributions of order statistics of n observations. They discussed statistical inference for parameters in the lifetimes of the components based on independent observations of lifetime of the system by using some characteristics of order statistics. Aki and Hirano (1996) showed a general method to derive the lifetime distribution of

a system whose components are assumed to be dependently distributed and proposed a practical dependence model to which the general method can be applied. Agin and Godbole (1992) and Koutras and Alexandrou (1997) developed testing problems for randomness based on success runs.

2 Some approaches to the distribution theory

Here, we overview some approaches to the study of discrete distribution theory. Of course, the standard method is enumerative combinatorics. That is to give all typical sequences. And, by splitting them into subsequences which can be interpreted. This method is useful when trials are independent or nearly independent (e.g. when the sequence of trials forms a Markov chain). However, when dependence of trials becomes strong (e.g. when the sequence of trials forms a higher order Markov chain), it becomes difficult even to write down all the typical sequences (Aki, Balakrishnan and Mohanty (1996)). There are further two major approaches to solving recent complicated problems. One is to give recurrence relations of the conditional probability generating functions of the desired random variable by considering the condition of one-step ahead from every condition. In some cases (e.g. when the system of recurrence relations is linear) the system of equations can be solved generally with respect to the conditional probability generating functions. Even if the system of equations can not be solved generally, in many cases the conditional probability generating functions can be written explicitly if the number of trials and the length of a run are specified (e.g. Ebneshahrashoob and Sobel (1990), Aki (1992), Aki and Hirano (1995), Uchida and Aki (1995), Aki, Balakrishnan and Mohanty (1996), Balakrishnan, Mohanty and Aki (1996), Hirano, Aki and Uchida (1997), Aki (1997), Han, Q. and Aki (1998, 2000a), Han, S.I. and Aki (2000), and Inoue and Aki (2001, 2002a)). The other effective approach is so called a Markov chain imbedding method, which is to imbed the sequence of trials into a Markov chain with an appropriate state

space. Then, the desired probability can be obtained by multiplying transition probability matrices (Fu and Koutras (1994b), Fu (1996), Koutras and Alexandrou (1995), Koutras (1997b), Koutras and Alexandrou (1997a,b), Han, Q. and Aki (1999, 2000b), Inoue and Aki (2002b) and Balakrishnan and Koutras (2002)).

Recently, Stefanov and Pakes (1997) presented a new approach for constructing joint generating function for quantities of interest associated with pattern formation in binary sequences. The method is based on exponential family of Markov chains.

Among the above approaches, the method of conditional generating functions seems to be useful for investigating more complicated situation in future. By developing the method, Aki (1999) derived the distribution of numbers of "1"-run of a specified length in a directed tree whose vertices are regarded to be $\{0, 1\}$ -valued random variables with a directed Markov distribution. The resulting system of equations of conditional probability generating functions is no longer linear if the directed tree is not a sequence. Hence, it may be difficult to apply the Markov chain imbedding method to the problem. Aki and Hirano (1999) treated sooner and later waiting time problems for runs in Markov dependent bivariate trials by using the method of conditional probability generating functions. The main purpose of this monograph is to show how to use the method of conditional probability generating functions.

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